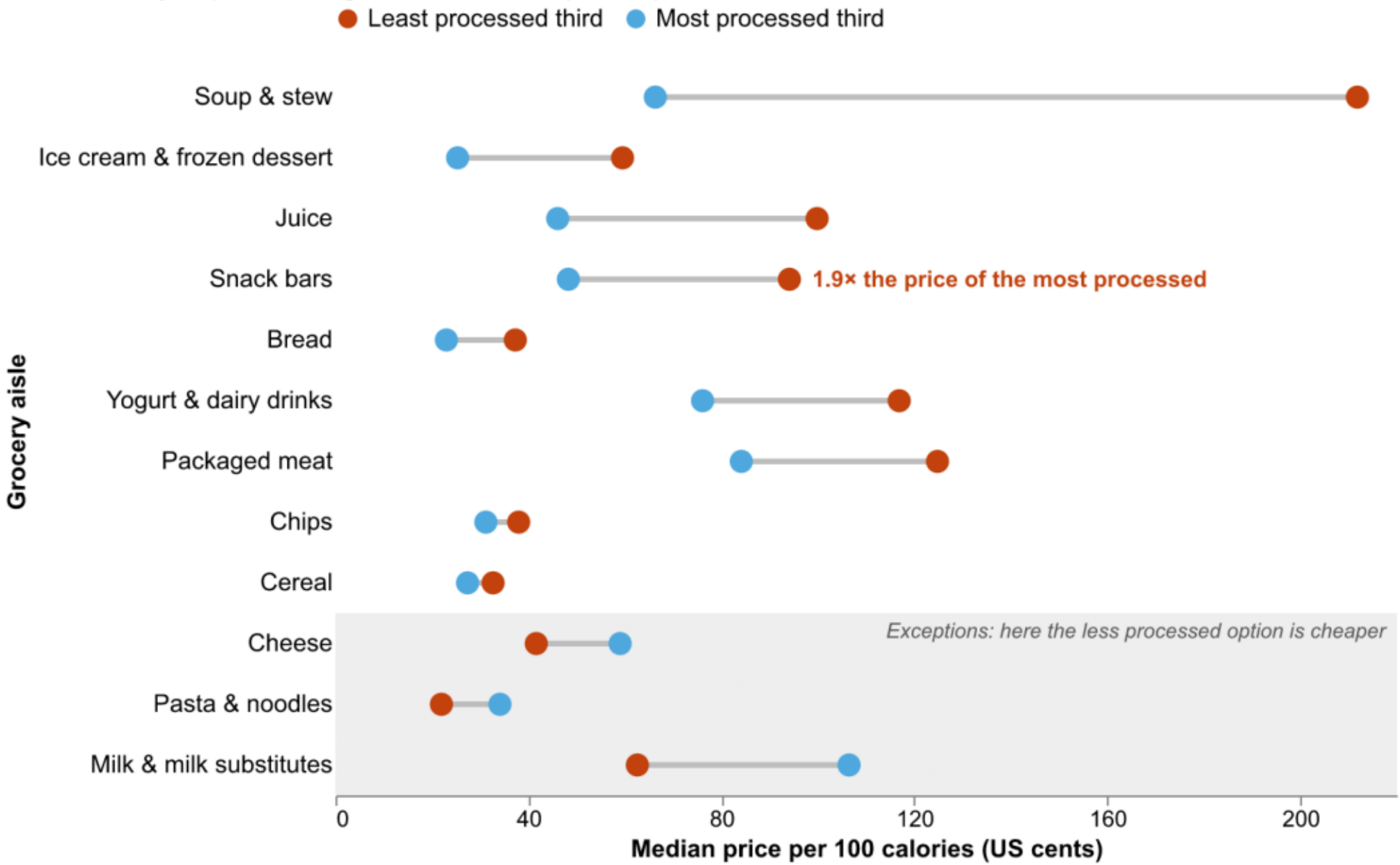
Less processed food costs more per calorie in 37 of 47 aisles

Products in each aisle are ranked by Food Processing Score (0 to 1, higher = more processed) and split into thirds.
Each dot is the median price of the top or bottom third, in US cents per 100 kcal.
Based on 19,645 products with a price and calorie count in the 47 aisles with 60+ such products.
Coffee beans excluded. 12 aisles shown. Per 100 g instead, the less processed third costs more in 34 of 47.
Source: GroceryDB (Walmart, Target and Whole Foods products).



Design rationale

Most people assume the less processed option is the healthier one, and that it costs more. I wanted to test the second half of that on real data. GroceryDB lists Walmart, Target and Whole Foods products with a Food Processing Score, a price and nutrition facts, so I asked: when two products sit in the same aisle, does the one that is less processed cost more? The chart answers yes in 37 of 47 aisles, and it shows how big the gap is in the aisles a shopper would recognize.

I kept the comparison inside each aisle because comparing across aisles would mix processing with product type (bread against steak). Within an aisle I ranked the priced products by Food Processing Score and took the median price of the least processed third and the most processed third. Price is converted to cents per 100 kcal so that a small bag of chips and a large carton of juice are measured on the same footing. Products without a price or a calorie count are dropped, aisles with fewer than 60 priced products are left out because a third of them is too small to trust, and coffee beans are excluded because they are listed at about zero calories. All of this is in the subtitle, along with the count of 19,645 products and the source. I used medians so a few luxury items cannot drag a third's price up.

Each aisle is one row, each third is one dot, and both dots sit on a shared horizontal price axis. Position along a common scale is the most accurate way to compare two values, and the gray line between the dots makes the gap itself the thing you read: a longer line means a bigger price difference. I chose dots over bars because the point is the difference between two values, not the size of either one, and bars would add ink without adding information. The axis starts at zero so the distances are not exaggerated. Rows are sorted by the size of the gap, so the eye moves from the biggest difference to the smallest.

For color I used orange for the less processed third and a calmer blue for the most processed third. Orange is the group the headline is about, so it carries the saturation, and the same orange is the accent in my other two checkpoint charts. Neither color is red or green, which avoids a good-versus-bad reading I cannot defend, and the pair differs along the blue-orange axis, which red-green color blindness leaves intact. The legend sits at the top so it is read before the data. I show 12 familiar aisles to keep the chart readable, and the subtitle gives the full count of 37 of 47 so the selection does not hide anything. The three aisles where the pattern flips (cheese, pasta and milk) are shaded and labeled instead of dropped. A single callout marks that snack bars cost 1.9 times as much when they are less processed, which gives the reader one concrete number to hold onto.

Two limits are worth knowing. Price per calorie favors calorie-dense food, so I repeated the comparison per 100 g. The pattern holds, though it weakens a little: the less processed third costs more in 34 of 47 aisles. It does change two familiar aisles. Soup, the largest gap per calorie, and yogurt are both cheaper per gram for the less processed third, because soup is mostly water and has few calories per gram. That is why I put the callout on snack bars, where the gap holds on both measures, rather than on the longest bar. The second limit is that a median hides the spread inside each third, and these are shelf prices from three stores, so the chart shows a pattern in the data and does not show that processing causes the price.